

IT Employee Lifecycle Automation System

Automate Onboarding & Offboarding. Zero Manual Steps, Zero Orphaned Accounts.

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Project
IT Employee
Lifecycle
Automation System

Demo Link
<http://localhost:3000>

Repository
[github.com/storrijas97/
Automated-User-
Onboarding](https://github.com/storrijas97/Automated-User-Onboarding)

⚠ The Problem

IT administrators spend hours manually creating Active Directory accounts for new hires — often *after* the employee's first day. Offboarding is worse: missed notifications leave terminated employees with active credentials, creating security vulnerabilities and compliance failures.

✓ The Solution

This system continuously monitors OrangeHRM for personnel changes and automatically executes the corresponding Active Directory actions. New hires get accounts on **Day 1**. Departing employees are disabled within **minutes of termination**. Every action is logged and auditable.

KEY FEATURES

1. OrangeHRM Sync

Polls OrangeHRM on a configurable schedule, keeping employee name, department, and job title current in real time.

2. Auto AD Provisioning

Creates AD accounts via LDAP with correct OU placement, display name, email, and enabled status on Day 1.

3. Role-Based Group Assignment

Maps department + job title combinations to AD security groups; assignments applied automatically at account creation.

4. Auto Deprovisioning

Detects terminations and immediately disables AD accounts, removing all group memberships to eliminate orphaned access.

5. Task Management

Tracks every operation (Pending → In Progress → Completed/Failed) with configurable retry logic and priority flagging.

6. Audit Log + CSV Export

Full chronological event log with date-range filtering and one-click CSV export for compliance and audit reviews.

7. Notification System

Real-time in-app notifications with unread badge counter, configurable via system settings.

8. Access Request Workflow

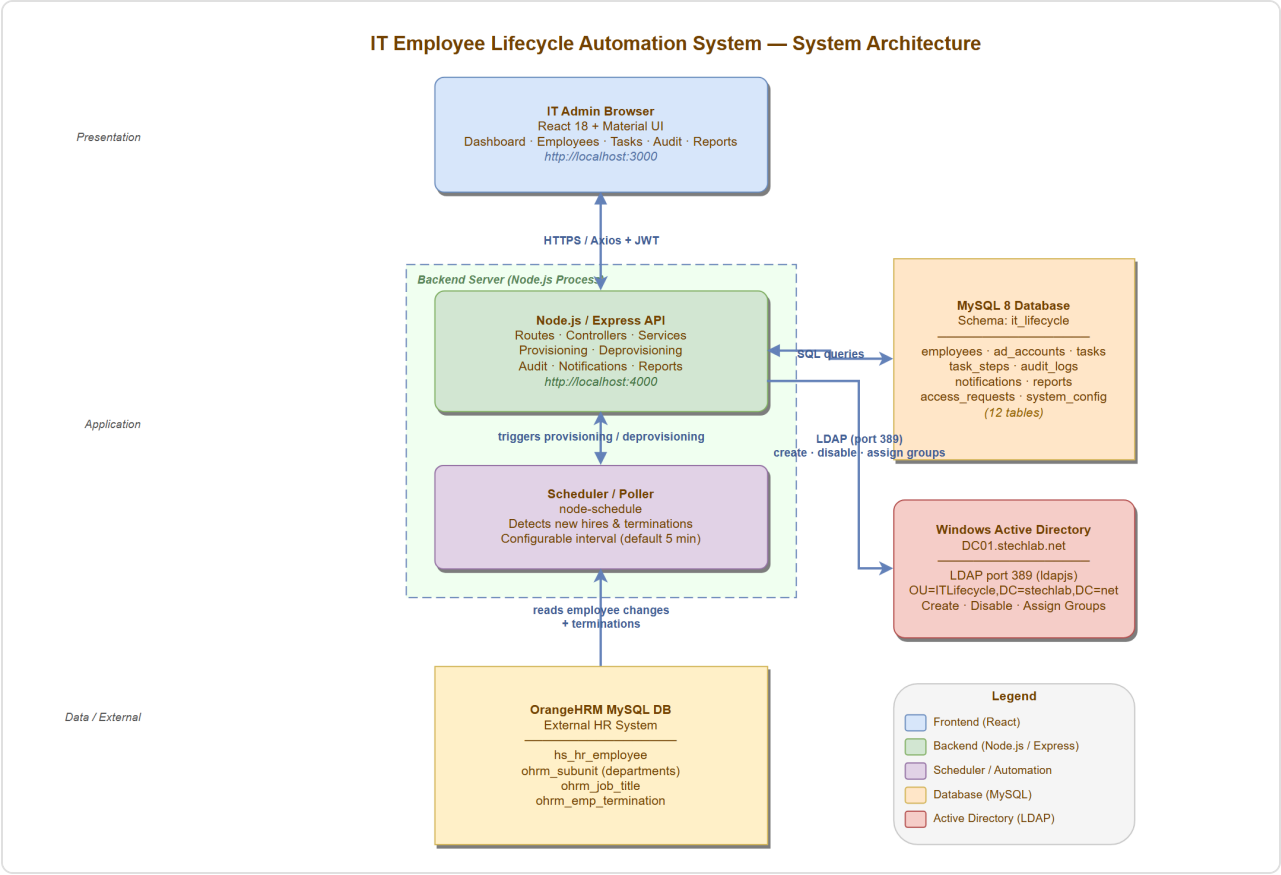
Managers submit additional-access requests. IT

	Admins approve or reject with a complete audit trail.
9. Compliance Reports Generate activity summaries filtered by date range and department, available for download as CSV for management or regulators.	10. System Configuration UI Adjust polling interval, retry limit, and notification settings at runtime with no code changes or restarts required.

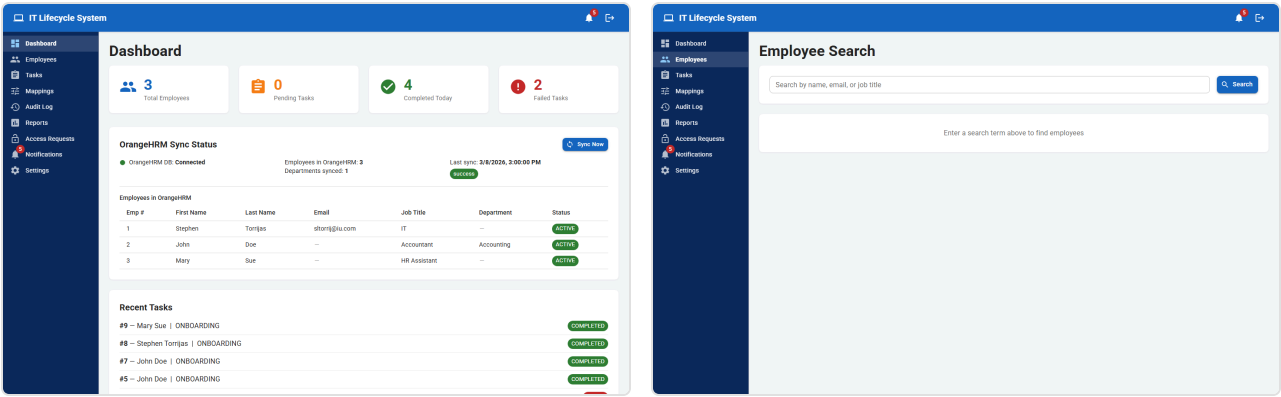
SYSTEM REQUIREMENTS

Component	Requirement
Runtime	Node.js 18+ (tested on v25.7.0)
Database	MySQL 8.0+ (schema: it_lifecycle, 12 tables)
HR System	OrangeHRM 5.x on XAMPP (direct MySQL DB access)
Directory	Windows Server Active Directory — LDAP port 389
OS	Windows Server 2019/2022 (for Active Directory integration)
Browser	Chrome 120+, Edge 120+, or Firefox 120+

SYSTEM ARCHITECTURE



SCREEN SNAPSHOTS



Dashboard: Task statistics & OrangeHRM sync status

Employee Search: AD account detail modal

IT Lifecycle System

Dashboard

Employees

Tasks

Mappings

Audit Log

Reports

Access Requests

Notifications

Settings

Tasks

Status

Type

Clear

#	Employee	Type	Status	Priority	Created	Actions
9	Mary Sue	ONBOARDING	COMPLETED	CRITICAL	3/8/2024, 1:39:31 PM	
8	Stephen Torrijas	ONBOARDING	COMPLETED	CRITICAL	3/8/2024, 1:23:33 PM	
7	John Doe	ONBOARDING	COMPLETED	CRITICAL	3/8/2024, 1:23:31 PM	
5	John Doe	ONBOARDING	COMPLETED	CRITICAL	3/8/2024, 1:05:57 PM	
6	Stephen Torrijas	ONBOARDING	FAILED	CRITICAL	3/8/2024, 9:22:41 PM	
4	John Doe	ONBOARDING	COMPLETED	CRITICAL	3/4/2024, 9:22:41 PM	
1	Stephen Torrijas	ONBOARDING	FAILED	CRITICAL	3/4/2024, 9:32:23 PM	

Task List: Status filters & priority flags

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Audit Log

Filter by actor

Filter by action

From date

mm/dd/yyyy

To date

mm/dd/yyyy

Apply Filters

Export CSV

Timestamp	Actor	Action	Target	Detail
3/8/2024, 1:39:31 PM	system	EMPLOYEE_PROVISIONED	marysue	Employee Mary Sue (2) provisioned
3/8/2024, 1:23:33 PM	system	EMPLOYEE_PROVISIONED	stephentorrijas	Employee Stephen Torrijas (1) provisioned
3/8/2024, 1:23:31 PM	system	EMPLOYEE_PROVISIONED	john doe	Employee John Doe (2) provisioned
3/8/2024, 1:05:01 PM	system	PROVISIONING_ERROR	1	creator failed: Unable to Perform
3/8/2024, 1:05:57 PM	system	EMPLOYEE_PROVISIONED	john doe	Employee John Doe (2) provisioned
3/8/2024, 9:22:41 PM	system	EMPLOYEE_PROVISIONED	john doe	Employee John Doe (2) provisioned
3/4/2024, 9:19:22 PM	system	PROVISIONING_ERROR	2	creator failed: Invalid Attribute Syntax
3/4/2024, 9:48:04 PM	system	PROVISIONING_ERROR	2	Invalid value found in an invalid LDAP attribute
3/4/2024, 9:32:29 PM	system	PROVISIONING_ERROR	1	name (string) required

1 - 9 of 9

Audit Log: Date range filtering & CSV export